

To solve, first evaluate the determinant of the coefficients, thus

$$D = \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

The first column of this determinant is formed from the coefficients of x and the second column contains the coefficients of y . Then, according to Cramer's rule, the solutions are:

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{D} \quad \text{and} \quad y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{D}.$$

Note that the determinant in the numerator of the equation for x is obtained from D by replacing the column of the coefficients of x with the constants on the right-hand side of the equations. Similarly y is given by the same equation but with the column of coefficients of y replaced by the right-hand constants. (You may recall using Cramer's rule to determine the currents in an electric circuit when applying Kirchhoff's laws.)

Unfortunately, when Cramer's rule is applied to *homogenous* linear equations such as [Equations \(11.31\)](#), the zeros on the right-hand side mean that the determinants in the numerators have a column of zeros. This leads to the *trivial solutions* $x = 0$ and $y = 0$. However, if the *denominator* is also zero (if $D = 0$), then $x = \frac{0}{0}$ and $y = \frac{0}{0}$, both of which are undefined. You may think this is not much of an advantage, but it actually allows you to find the solution! In fact it is a general rule that coupled homogeneous algebraic equations have nontrivial solutions *if and only if* the determinant of the coefficients is zero.

Going back to [Equations \(11.31\)](#), the condition for a nontrivial solution is that the determinant of the coefficients is equal to zero:

$$D = \begin{vmatrix} -\omega^2 + \omega_0^2 & -\Delta\omega^2 \\ -\Delta\omega^2 & -\omega^2 + \omega_0^2 \end{vmatrix} = 0,$$

or

$$\begin{aligned} (-\omega^2 + \omega_0^2)^2 - (\Delta\omega^2)^2 &= 0, \\ \omega^2 &= +\omega_0^2 \mp \Delta\omega^2. \end{aligned}$$

That is,

$$\omega = \pm(\omega_0^2 \mp \Delta\omega^2)^{\frac{1}{2}}.$$

This yields four possible values for ω . Two of them are

$$\omega_1 = \pm(\omega_0^2 - \Delta\omega^2)^{\frac{1}{2}},$$

and the other two are

$$\omega_2 = \pm(\omega_0^2 + \Delta\omega^2)^{\frac{1}{2}}. \quad (11.32)$$

The general solutions for x_1 and x_2 are the sum of all possible solutions:

$$\begin{aligned} x_1 &= A_1 e^{i\omega_1 t} + A_{-1} e^{-i\omega_1 t} + A_2 e^{i\omega_2 t} + A_{-2} e^{-i\omega_2 t}, \\ x_2 &= B_1 e^{i\omega_1 t} + B_{-1} e^{-i\omega_1 t} + B_2 e^{i\omega_2 t} + B_{-2} e^{-i\omega_2 t}. \end{aligned} \quad (11.33)$$

These two expressions involve eight constants. But solving two second-order differential equations should lead to only four constants. Therefore, not all of the eight constants are independent. We now determine the four independent constant amplitudes.

Note that each term in our expression for x_1 satisfies the differential equation. Thus, for example, a possible solution involves oscillations of frequency ω_1 . Then,

13 Waves

A wave is an oscillation of a medium, such as a water wave in the ocean or a wave propagating down a stretched slinky or a taut string. In general, the medium as a whole does not translate. For example, a wave in the ocean can be thought of as water molecules moving vertically up and down while the wave itself moves horizontally. A strange and interesting thing about waves is that although there is no transport of mass, the wave does transport energy and momentum.

A *transverse* wave, such as a wave in a string, is one in which the material particles move perpendicular to the direction of the wave. On the other hand, a sound wave in air or the wave set up in a metal rod when you hit one end with a hammer, are examples of *longitudinal* waves. In these waves the material particles move back and forth parallel to the direction of the wave.

This chapter is an introduction to wave motion and is limited to an analysis of waves in strings.² However, the wave equation and the wave properties derived here are valid for all types of waves. The mathematical techniques considered in this chapter include expressing a periodic function as a Fourier series and solving the wave equation by separation of variables. You will find the material of this chapter to be particularly useful when you study electromagnetic waves.

13.1 A Wave in a Stretched String

Consider a long string. If you shake one end of the string up and down, you can set up a *traveling wave*. A single shake will generate a pulse that moves along the string; continuous shaking will produce a wave train which is often sinusoidal in form. If you have a shorter string, such as one of the strings in your guitar, and you pluck the string, you can set up a *standing wave*.

A wave is characterized by a number of physical parameters. The main ones are the following.

Amplitude (A): The maximum displacement from equilibrium.

Wavelength (λ): The distance from peak to peak or between any two corresponding points on the wave.

Period (τ): The time it takes for a point on the wave to go through one complete oscillation.

Frequency (f): The number of oscillations per unit time. Note that $f = 1/\tau$.

Speed (v): How fast the waveform is displaced in the direction of its motion, $v = \lambda/\tau = \lambda f$.

Angular frequency (ω): Another measure of frequency, defined by $\omega = 2\pi f = 2\pi/\tau$.

¹ This is not quite true. A molecule of water also has a small horizontal oscillation and traces out an elliptical path – but you get the general idea.

² An ideal string is perfectly flexible and linearly elastic. This means the tension is everywhere the same and always directed tangentially. Linear elasticity means the tension depends linearly on the amount the string is stretched (Hooke's law).

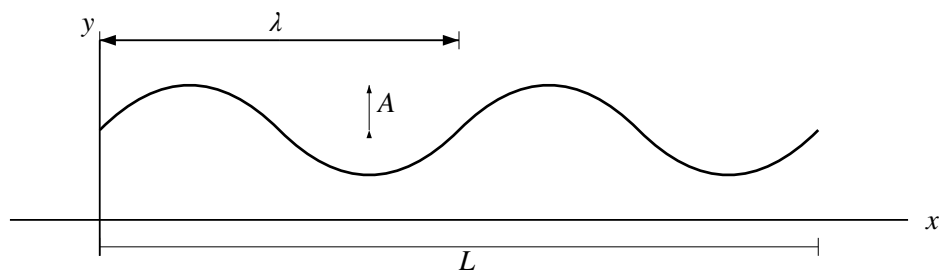


Figure 13.1 A wave in a string of length L . The wavelength and amplitude are indicated on the figure. The displacement from equilibrium is $y = y(x)$.

Wave number (k): A parameter related to wavelength, defined by $k = 2\pi/\lambda$.

Because $\lambda = v\tau$ the wave number can also be expressed as $k = \omega/v$. Note that the wave number is analogous to the angular frequency: one of them is an inverse time and the other is an inverse length.

Consider a string of length L with fixed end points. Suppose you generated a wave in the string and took a photograph of the wave. In your photograph the string might look somewhat like the sketch in [Figure 13.1](#). The vertical displacement of a particle in the string is y . This displacement depends on the horizontal position, x , so $y = y(x)$. (There is no time dependence yet because this is just a description of the instantaneous photograph of the wave.) The curve repeats after a distance $x = \lambda$. Suppose you measured the vertical displacement (y) along the string and found it satisfied the following sinusoidal curve³

$$y(x) = A \sin\left(\frac{2\pi x}{\lambda}\right) = A \sin(kx).$$

When x is any multiple of $\lambda/2$ the quantity in parentheses is a multiple of π and there is zero displacement at that point. Such points are called the *nodes* of the waveform.

The wave in [Figure 13.1](#) has a wavelength equal to one half the length of the string, L . The only waves that will “fit” onto the string are those with wavelengths $\lambda = 2L/n$ where $n = 1, 2, 3, \dots$

Now imagine that instead of an instantaneous photograph you have a video of the oscillating string. If you look at a particular point on the string you will see it moving up and down. Thus the displacement y is also a function of the time t . As you might suspect, the wave can be expressed mathematically by an expression involving sines and cosines of the quantities kx and ωt .

To derive the expression for $y = y(x, t)$ we need to determine the equation of motion and then integrate it. Consider a string of length L under tension F . The two ends of the string are at $x = 0$ and $x = L$. Ignore gravity. If you pluck the string

you can set up a “standing wave” as in a guitar string. If you look at a plucked guitar string you will notice that the amplitude of the wave is much smaller than the length of the string. That is, y is relatively a small quantity. Furthermore, the slope dy/dx is also small. The tension in the string is the same at every point.

Figure 13.2 shows an infinitesimal portion (dx) of the string as a pulse propagates along it. The portion of the string extends from x to $x + dx$. The mass of this element is $dm = \rho dx$, where ρ is the mass per unit length. The two ends are displaced vertically by $y(x, t)$ and $y(x + dx, t)$. The element of the string is subjected to a tension $F(x)$ acting to the left and down at x (at angle θ) and a tension $F(x + dx)$ acting to the right and up at $x + dx$ (at angle ϕ). The horizontal components of these two tensions must cancel because the element of string is not accelerating to...

³ As you will see, a wave can generally be expressed as a sum of sines and cosines. For simplicity, at the moment, we assume a single sinusoid.

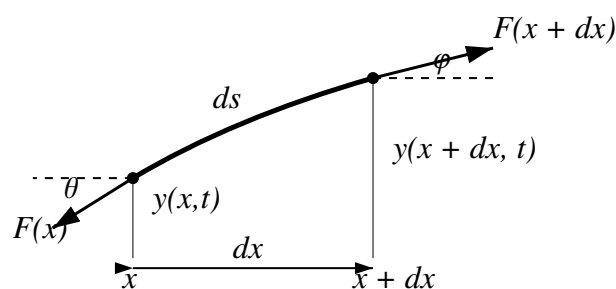


Figure 13.2 A small portion of a vibrating string.

the right or the left. (The tension in the string is the same at all points so there actually is a small oscillating horizontal force that we shall consider in [Section 13.7](#).) There is, however, a vertical component of force bringing the string back down to its equilibrium position.

The magnitude of the tension is the same at both ends, that is,

$$F(x + dx) = F(x) = F.$$

The vertical acceleration is $a_y = \partial^2 y / \partial t^2$, so applying Newton's second law gives⁴

$$\begin{aligned}\rho dx \frac{\partial^2 y(x, t)}{\partial t^2} &= F(x + dx) \sin \phi - F(x) \sin \theta, \\ &= F(\sin \phi - \sin \theta).\end{aligned}$$

Assuming the angles θ and ϕ are small (because both y and dy/dx are small)

$$\begin{aligned}\sin \theta &\doteq \tan \theta = \frac{\partial y(x, t)}{\partial x}, \\ \sin \phi &\doteq \tan \phi = \frac{\partial y(x + dx, t)}{\partial x}.\end{aligned}$$

So,

$$\rho dx \frac{\partial^2 y(x, t)}{\partial t^2} = F \left[\frac{\partial y(x + dx, t)}{\partial x} - \frac{\partial y(x, t)}{\partial x} \right].$$

Dividing both sides by dx and taking the limit as $dx \rightarrow 0$ leads to

$$\rho \frac{\partial^2 y}{\partial t^2} = F \frac{\partial}{\partial x} \left[\frac{\partial y}{\partial x} \right],$$

or

$$\frac{\partial^2 y}{\partial x^2} = \frac{\rho}{F} \frac{\partial^2 y}{\partial t^2}.$$

If this last equation looks familiar to you it is because it has the form of the well-known relation called the “wave equation” which is usually expressed as

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}, \quad (13.1)$$

where v is the speed of the wave.

⁴ We use partial derivatives because the acceleration depends on x as well as t .

Thus we have not only obtained the equation of motion for the wave in a taut string, we have also shown that the wave propagates in the x direction with speed

$$v = \sqrt{F/\rho}.$$

The wave equation (13.1) is a very important, fundamental relation. It is a second-order partial differential equation giving the displacement of the string as a function of time and position. I strongly suggest you memorize it.

Exercise 13.1

A string of length 6 m is fixed at both ends. Its mass is 0.1 kg and the tension in the string is 50 N. (a) What is the wavelength of the longest possible standing wave in this string? (b) What is the frequency of that wave? Answer: (a) 12 m, (b) 4.56 Hz.

Exercise 13.2

If n is the number of half wavelengths, show that the frequency of a standing wave in a string is $f = nv/2L$.

13.2 Direct Solution of the Wave Equation

Our next task is to solve Equation (13.1), the wave equation, to obtain an expression for $y = y(x, t)$. Many second-order partial differential equations (including this one) can be solved by a technique called *separation of variables*.

The basic idea behind the separation of variables method is to express the unknown function (in our case $y = y(x, t)$) as the product of two functions, each depending on only one variable. That is, you assume that you can write $y(x, t)$ in the following form:

$$y(x, t) = X(x)T(t),$$

where X is a function only of x and T is a function only of t . Substitute this expression into the differential equation (13.1) to get

$$\begin{aligned} \frac{\partial^2 X(x)T(t)}{\partial t^2} &= v^2 \frac{\partial^2 X(x)T(t)}{\partial x^2}, \\ X \frac{\partial^2 T}{\partial t^2} &= v^2 T \frac{\partial^2 X}{\partial x^2}, \end{aligned}$$

$$\frac{1}{T} \frac{\partial^2 T}{\partial t^2} = \frac{v^2}{X} \frac{\partial^2 X}{\partial x^2}.$$

Now the left-hand side depends only on t and the right-hand side depends only on x . Since x and t can be varied independently, the only way the equation can be satisfied is if the right-hand side and the left-hand side are equal to the same constant. Call it $-\omega^2$. The partial differential equation has then been *separated* into the following two *ordinary* differential equations:

$$\begin{aligned} \frac{v^2}{X} \frac{d^2 X}{dx^2} &= -\omega^2, \\ \frac{1}{T} \frac{d^2 T}{dt^2} &= -\omega^2, \end{aligned}$$

or

$$\begin{aligned} \frac{d^2 X}{dx^2} + \frac{\omega^2}{v^2} X &= 0, \\ \frac{d^2 T}{dt^2} + \omega^2 T &= 0. \end{aligned}$$

Both of these equations have the form of the simple harmonic oscillator equation. You have seen the SHM equation many times, so I will simply write down the general solutions

$$\begin{aligned} X(x) &= A \cos \frac{\omega}{v} x + B \sin \frac{\omega}{v} x = A \cos kx + B \sin kx, \\ T(t) &= C \cos \omega t + D \sin \omega t. \end{aligned}$$

The solution for $y = y(x, t)$ is the product of these two functions, thus:

$$y(x, t) = X(x)T(t) = (A \cos kx + B \sin kx)(C \cos \omega t + D \sin \omega t). \quad (13.2)$$

Note that the partial differential equation (13.1) is satisfied by the four expressions

$$y(x, t) \propto \begin{cases} \cos kx \cos \omega t \\ \cos kx \sin \omega t \\ \sin kx \cos \omega t \\ \sin kx \sin \omega t \end{cases}.$$

The general solution, [Equation \(13.2\)](#), is the sum of all of these. The coefficients,

A, B, C, D are determined from the boundary conditions.

Exercise 13.3

Show that an expression of the form

$$y = A \sin kx + B \cos kx$$

can be expressed as

$$y = C \cos(kx + \alpha).$$

Express C and α in terms of A and B . (The quantity α is called the “phase constant.”) Answer: $C = \sqrt{A^2 + B^2}$, $\alpha = \tan^{-1}(-A/B)$.

13.2.1 Fourier Series (Optional)

You are probably familiar with Fourier series from your mathematics courses, so this discussion is intended to refresh your memory. If you have not been exposed to Fourier series previously, I strongly suggest you read about them in a “Math Methods” text.⁵

Any periodic function (such as a wave) or a portion of a periodic function (such as a standing wave) can be represented as a sum of sines and cosines. Consider, for example, the portion of a “square wave” shown in [Figure 13.3](#).

The function we are approximating with a Fourier series is $f(x) = -\pi/2$ for $-\pi < x < 0$, $f(x) = +\pi/2$ for $0 < x < +\pi$. This is a portion of the square wave which alternates between plus and minus $\pi/2$ with a periodicity of π . In the middle panel of [Figure 13.3](#), I plotted $2 \sin x$

⁵ An excellent source is [Chapter 7](#) in *Mathematical Methods in the Physical Sciences* by Mary L. Boas (Wiley and Sons, 2006).

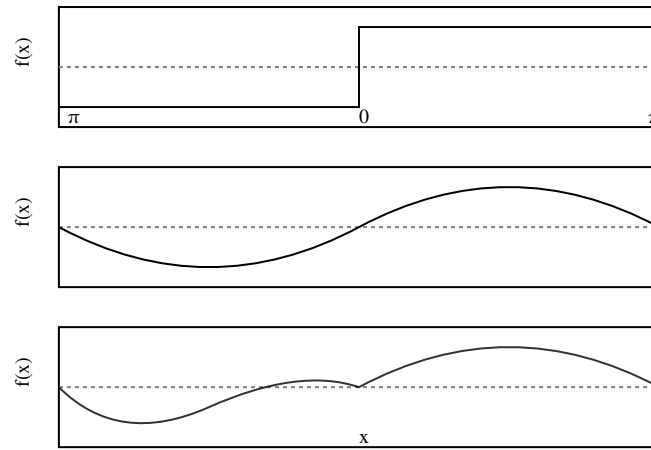


Figure 13.3 The top panel shows the square wave... The middle panel also shows $f(x) = 2 \sin x$... The bottom panel shows $f(x) = 2 \sin x + \frac{2}{3} \sin 3x$...

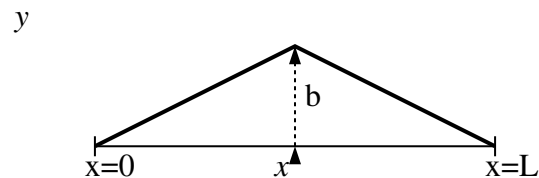


Figure 13.4 A string plucked at its center. The displacement at $L/2$ is b .

which has the appropriate periodicity but does not approximate the original function very well. In the bottom panel the additional term $\frac{2}{3} \sin 3x$ has been added to our highly truncated Fourier series, and you can appreciate that the approximation is getting better. The next approximation (not plotted) is $f(x) = 2 \sin x + \frac{2}{3} \sin 3x + \frac{2}{5} \sin 5x$. The Fourier series more and more closely approximates the square wave as more and more terms are included.

Before getting into the details of how to obtain a Fourier series, let us consider why they are useful. That is, if we already know the function $f(x)$, why would we want to express it as an infinite sum of sines and cosines? To answer this question let us note that the solution to the wave equation (13.2) is expressed in terms of sines and cosines which are multiplied by constant coefficients which we need to determine. These coefficients will be determined from the initial and/or boundary conditions. In the problem of an oscillating mass on a spring or a pendulum this was easily done because the initial conditions were the position and the velocity of what was essentially a point mass. But the initial conditions for a vibrating string involve the positions and velocities of all the points of the string. Suppose that we know the initial configuration of a plucked string, and it looks like Figure 13.4, which shows two straight lines. How can we match this initial condition to the solution that looks like Equation (13.2), i.e., a sum of sines and cosines? The answer is to express such a function as that of Figure 13.4 as a Fourier series and thus determine the coefficients in the solution.

In general a function $f(x)$ defined in the interval $-\pi < x < +\pi$ can be expressed as a Fourier series having the form

$$f(x) = \frac{1}{2}a_0 + a_1 \cos x + a_2 \cos 2x + \cdots + b_1 \sin x + b_2 \sin(2x) + \cdots$$

or

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx. \quad (13.3)$$

The problem now reduces to determining the coefficients a_n and b_n . This is accomplished by using the orthogonality properties of sines and cosines which can be expressed by the following three integrals:

$$\begin{aligned} \int_{-\pi}^{+\pi} \sin nx \cos mx \, dx &= 0 \text{ for all } m \text{ and } n \\ \int_{-\pi}^{+\pi} \sin nx \sin mx \, dx &= \pi \text{ for } n = m, \text{ and } 0 \text{ for } n \neq m \\ \int_{-\pi}^{+\pi} \cos nx \cos mx \, dx &= \pi \text{ for } n = m, \text{ and } 0 \text{ for } n \neq m \end{aligned}$$

To determine the coefficients a_n and b_n we multiply Equation (13.3) by $\cos mx$ and integrate from $-\pi$ to $+\pi$:

$$\begin{aligned} \int_{-\pi}^{+\pi} f(x) \cos mx \, dx &= \frac{a_0}{2} \int_{-\pi}^{+\pi} \cos mx \, dx \\ &+ \int_{-\pi}^{+\pi} \sum_{n=1}^{\infty} a_n \cos nx \cos mx \, dx \\ &+ \int_{-\pi}^{+\pi} \sum_{n=1}^{\infty} b_n \sin nx \cos mx \, dx. \end{aligned}$$

For $m = 0$ this reduces to $\int_{-\pi}^{+\pi} f(x) dx = \int_{-\pi}^{+\pi} \frac{a_0}{2} 1 dx = \frac{a_0}{2}(\pi + \pi)$, or

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) dx$$

and for $m \neq 0$

$$\begin{aligned}\int_{-\pi}^{+\pi} f(x) \cos mx \, dx &= \frac{a_0}{2} \int_{-\pi}^{+\pi} \cos mx \, dx + \sum_{n=1}^{\infty} a_n \pi \delta_{mn} + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{+\pi} \sin nx \cos mx \, dx \\ &= 0 + a_m \pi + 0.\end{aligned}$$

Therefore,

$$a_m = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) \cos mx \, dx.$$

Similarly, if we multiply [Equation \(13.3\)](#) by $\sin mx$ and integrate we obtain

$$b_m = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) \sin mx \, dx.$$

We assumed the function $f(x)$ was defined in the interval $-\pi$ to $+\pi$. But this assumption was only made so that we could keep the notation simple. For a function defined on the arbitrary interval $-l$ to $+l$ we simply change the variable from x to $\frac{\pi x}{l}$ and obtain the coefficients from

$$a_m = \frac{1}{l} \int_{-l}^{+l} f(x) \cos \frac{m\pi}{l} x \, dx \quad m = 0, 1, 2, 3, \dots$$

$$b_m = \frac{1}{l} \int_{-l}^{+l} f(x) \sin \frac{m\pi}{l} x \, dx \quad m = 1, 2, 3, \dots$$

A labor saving fact is that if $f(x)$ is an *odd* function of x ($f(x) = -f(-x)$), then only sine terms appear in the Fourier series, and if $f(x)$ is an *even* function of x ($f(x) = f(-x)$), then only cosine terms appear in the series.

Exercise 13.4

Show that if $f(x)$ is defined in the interval $0 < x < l$, the coefficients in the Fourier expansion are

$$a_m = \frac{2}{l} \int_0^l f(x) \cos \frac{m\pi}{l} x dx,$$
$$b_m = \frac{2}{l} \int_0^l f(x) \sin \frac{m\pi}{l} x dx.$$

13.3 Standing Waves

As an example of how to determine the coefficients in the general solution, [Equation \(13.2\)](#), consider a standing wave, as in the string illustrated in [Figure 13.1](#). The string is tied down at the end points so

$$y(x=0) = y(x=L) = 0.$$

Then

$$y(0, t) = 0 = (A \cos 0 + B \sin 0)(C \cos \omega t + D \sin \omega t),$$

or

$$0 = A(C \cos \omega t + D \sin \omega t).$$

This requires that $A = 0$. The general equation has now been reduced to

$$y(x, t) = B \sin(kx)(C \cos \omega t + D \sin \omega t).$$

The boundary condition at $x = L$ leads to

$$y(L, t) = 0 = B \sin(kL)(C \cos \omega t + D \sin \omega t).$$

This condition is satisfied if $B = 0$, but in that case $y = 0$ everywhere and at all times! Although it satisfies the differential equation, this is obviously not a satisfactory solution. (It is the “trivial” solution.) However, there is another way to ensure that this expression is zero; if

$$\sin(kL) = 0,$$

the condition is satisfied. This means that kL must be an integer multiple of π because if $\sin \theta = 0$, then $\theta = 0, \pi, 2\pi, 3\pi, \dots$. That is,

$$kL = n\pi, \quad n = 1, 2, 3, \dots$$

Note that $n = 0$ is not included because that would require $k = 0$ and again $y(x, t)$

would be zero for all values of x . If k can take on the values $n\pi/L$, we must also require that ω take on a set of allowed values because $\omega = kv = k\sqrt{F/\rho}$. Thus, ω has the values

$$\omega_n = \frac{n\pi}{L} \sqrt{\frac{F}{\rho}} = \frac{n\pi v}{L},$$

and Equation (13.2) reduces to

$$y(x, t) = B \sin \frac{n\pi x}{L} (C \cos \omega_n t + D \sin \omega_n t).$$

Finally, combining the constants BC and BD and calling them A_n and B_n we write

$$y(x, t) = (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{L}. \quad (13.4)$$

The differential equation and the boundary conditions are satisfied for any integer value of n . But note that the coefficients A_n and B_n are generally different for different values of n . The general solution to the differential equation is, of course, the sum of all possible solutions, that is,

$$\begin{aligned} y(x, t) &= \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{L}, \\ &= \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi v}{L} t + B_n \sin \frac{n\pi v}{L} t \right) \sin \frac{n\pi x}{L}. \end{aligned} \quad (13.5)$$

To determine the coefficients A_n and B_n we use the initial conditions, as illustrated in the following example.

Worked Example 13.1 A string of length L is pulled up at its midpoint a distance b and then released. The displacement of the string at any position ($0 \leq x \leq L$) and time ($t \geq 0$) is given by Equation (13.5). Determine the coefficients A_n and B_n . See Figure 13.4.

Solution The waveform at $t = 0$ is given by the two straight lines,
 $y = (2b/L)x$ for $0 \leq x \leq L/2$ and $y = (-2b/L)x + 2b$ for $L/2 \leq x \leq L$.

At time $t = 0$ the string is not moving, so $\partial y / \partial t = 0$. From [Equation \(13.5\)](#)

$$0 = \frac{\partial}{\partial t} \left[\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L} \cos \omega_n t + \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} \sin \omega_n t \right]_{t=0},$$

$$0 = \left[\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L} (-\omega_n \sin \omega_n t) + \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} (\omega_n \cos \omega_n t) \right]_{t=0},$$

$$0 = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}.$$

This is zero for all values of x iff $B_n = 0$ for all n . So we are left with

$$y(x, t) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L} \cos \omega_n t,$$

$$y(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L}.$$

But $y(x, 0)$ has the shape illustrated in [Figure 13.4](#). As you will show in [Problem 13.1](#), this can be expressed as the following Fourier series:

$$y(x, 0) = \frac{8b}{\pi^2} \left(\sin \frac{\pi x}{L} - \frac{1}{3^2} \sin \frac{3\pi x}{L} + \frac{1}{5^2} \sin \frac{5\pi x}{L} - \cdots \right). \quad (13.6)$$

That is,

$$y(x, 0) = \frac{8b}{\pi^2} \left(\sin \frac{\pi x}{L} - \frac{1}{3^2} \sin \frac{3\pi x}{L} + \frac{1}{5^2} \sin \frac{5\pi x}{L} - \cdots \right)$$

$$= \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L},$$